

Portfolio Question 2.2.P - Leave-one-out Cross Validation

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Portfolio 2.2.P (v4.0.0)

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- ▶ Consider cross validation where we leave out only one datapoint. Is this a good idea? Is it computable for large data?
 - ▶ Relate the maths to linear regression.
 - ▶ How does leave-one-out relate to other forms of cross validation?

LOOCV

- ▶ We write a statistic $\hat{s}$ based on all data $\{y\}$ except i as $\hat{s}^{(-i)}$ and the data is $\{y\}^{(-i)}$.
- ▶ For a general **loss function** we can write:

$$LOOCV = \frac{1}{n} \sum_{i=1}^n \text{Loss} \left(y_i; \hat{\theta} | y^{(-i)} \right)$$

- ▶ i.e. we evaluate the loss function for each datapoint using the estimate from the remaining data.
- ▶ NB **A loss function** is something that we choose the parameters θ to minimise. It can be:
 - ▶ the MSE,
 - ▶ the (negative log) likelihood,
 - ▶ a penalised version of these,
 - ▶ or any other convenient quantity.

LOOCV for linear models

- ▶ For the MSE of a linear model we can write:

$$LOOCV = \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{y}_i^{(-i)} \right)^2$$

- ▶ It is not particularly straightforward¹ to show that:

$$LOOCV = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - H_{ii}} \right)^2$$

- ▶ Where H is a function of X only (see Appendix).
- ▶ This is a very important quantity, often called the **Studentized residual**
- ▶ i.e. the LOOCV can be directly computed from a regression containing all data:
 - ▶ “downweighting” low-leverage data
 - ▶ “upweighting” high-leverage (hard to predict) data.

¹Our references avoid proving this, but do discuss the motivation. **Proofs are available.**

Leave-one-out Cross-Validation

- ▶ Leaving out a single datapoint is going to be insufficient unless the **data are independent**.
- ▶ The real world is rarely completely independent.
- ▶ However, if its computationally convenient to compute LOOCV it is better than leaving nothing out.
- ▶ Analogous tricks work for:
 - ▶ **Linear models** including **Best Linear Unbiased Predictors** (BLUPs)
 - ▶ **Kernel methods**
 - ▶ **Nearest neighbour** methods
 - ▶ And others

Asymptotics

- ▶ Some interesting facts about LOOCV?
 - ▶ As $n \rightarrow \infty$, the expected out-of-sample MSE of the model picked by LOO cross-validation is **close to that of the best model** considered.
 - ▶ As $n \rightarrow \infty$, if the true model is among those being compared, LOOCV tends to pick a **strictly larger model** than the truth.
- ▶ LOOCV is not the right tool for choosing the **right model**.
- ▶ It is however an excellent tool for choosing the model with the best out-of-sample **predictive power**.

Discussion

- ▶ If data are correlated, is leaving out one datapoint at a time enough?
- ▶ How do we compute it? What if there isn't a trick way to do it efficiently?
- ▶ Does it matter that each point uses the same $n - 2$ data as each other point?

Appendix: The Hat Matrix

- ▶ There is an important and **response independent** quantity hidden in the prediction:

$$H = X(X^T X)^{-1} X^T$$

- ▶ The fitted values are:

$$\hat{\mathbf{y}} = X\hat{\boldsymbol{\beta}} = X(X^T X)^{-1} X^T \mathbf{y} = H\mathbf{y}$$

- ▶ H is dimension $N \times N$
- ▶ H “projects” $\mathbf{y}$ into the fitted value space $\hat{\mathbf{y}}$
- ▶ Put the “hat” on y

Appendix: Properties of the Hat Matrix

- ▶ **Influence:** $\frac{\partial \hat{y}_i}{\partial y_j} = H_{ij}$. So H controls how much a change in one observation changes the estimates of each other point.
- ▶ **symmetry:** $H^T = H$. So influence is symmetric.
- ▶ **Idempotency:** $H^2 = H$. So the predicted value for any projected point is the predicted value itself.
- ▶ You should read up on these and other vector algebra properties.

Appendix: Residuals and the Hat Matrix

- ▶ The residuals can be written:

$$\mathbf{e} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\mathbf{y}$$

- ▶ $\mathbf{I} - \mathbf{H}$ is also symmetric and idempotent, and can also be interpreted in terms of Influence.
- ▶ Because of this,

$$\text{MSE}(\hat{\beta}) = \frac{1}{n} \mathbf{y}^T (\mathbf{I} - \mathbf{H})^T (\mathbf{I} - \mathbf{H}) \mathbf{y} = \frac{1}{n} \mathbf{y}^T (\mathbf{I} - \mathbf{H}) \mathbf{y}$$